BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES

COURSE HANDOUT

Part A: Content Design

Course Title	**Full Stack Application Development**
Course No(s)	SE* ZG503
Credit Units	4
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Version	1.0
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Course Description:

The modern software application landscape is evolving rapidly, moving from the conventional, layered web applications hosted on remote servers to the mobile-only / mobile-first and cloud-native applications with complex deployment options. At the same time, the software development teams have started adapting the most agile development methodologies, enabling them to deliver the software with better quality at shorter and more frequent intervals. These developments have resulted in the necessity of engineers having multiple skill sets essential for every aspect of the software development life cycle, right from requirement analysis to application deployment.

This course aims to provide a comprehensive introduction to modern Web application architecture approaches, frontend and backend technologies, and suitable web application frameworks required for developing modern web apps. It focuses on designing and implementing end-to-end functional web applications, learning the key patterns followed at each layer of the application architecture and technology considerations to choose an appropriate implementation technology. The course also involves hands-on exposure to full-stack development of web applications using development frameworks.

Course Objectives

No	Course Objective

CO1	Understand the modern application landscape and the evolution of the application landscape.
CO2	Build an understanding of an end-to-end application's typical structure, design, and implementation considerations.
CO3	Comprehend the necessity and usefulness of the client and server-side frameworks, along with their strengths and weaknesses.
CO4	Develop and test a working model of web application using the tech stack.

Text Book(s)

T1	Web Development with Node and Express by Ethan Brown , Oreilly Media 2nd Edition , 2019

Reference Book(s) & other resources

R1	Building Microservices: Designing Fine-Grained Systems Book by Sam Newman, 1st edition, published by O'Reilly Media, Feb 2015
R2	The Design of Web APIs by Arnaud Lauret Published by Manning Publications; 1st edition (November 2019)
R3	Full Stack Web Development: The Comprehensive Guide by Philip Ackermann Shroff/Rheinwerk Computing; First Edition (2 August 2023)
R4	Microservice APIs: Using Python, Flask, FastAPI, OpenAPI and More. Peralta, J. H. (2023): Manning.
R5	GRPC: Up and Running: Building Cloud Native Applications with Go and Java for Docker and Kubernetes Book by Danesh Kuruppu and Kasun Indrasir, Oreilly Media
Web Resources	Mozilla Developer Network https://developer.mozilla.org/en-US/ React JS https://react.dev/learn

<u>Content Structure</u>

Module 1: Application Development

- Introduction to the various Application Landscape: [Web applications, Mobile applications, Cross Platform applications, Cloud native applications, Serverless Applications]
- Layered Architectures (client /server, 2/3 tier- N tier)
- Monolithic
- Distributed Architectures – Service-oriented architecture, Microservices
- MVC Pattern

Module 2: Understanding the Basics

- Structure of web applications
    - Frontend (HTML, CSS, JavaScript),
    - Backend Server Side logic, API, Web Services
    - Database
- Client – Server Communication
- Relationship between URLs, Domains, and IP Addresses
- Domain Name Systems, Content Delivery Networks
- Technologies and Tools for Full stack development

Module 3: Web Protocols

- HTTP
    - HTTP Request- Response and its structure
    - HTTP Methods;
    - HTTP Headers
    - Connection management - HTTP/1.1 and HTTP/2
- Synchronous and asynchronous communication
- Communication with Backend
    - AJAX, Fetch API
    - Webhooks
    - Server-Sent Events
- Polling
- Bidirectional communication - Web sockets

Module 4: Server Side: Implementing Web Services

- REST
    - Principles of REST
    - REST constraints
    - Service Design with REST
        - Interaction Design with HTTP
        - Interface Design (URI)
        - Representation and Metadata design
    - Implementing REST API
        - Using a Framework
        - URL Mapping
        - Routing Requests-Redirection
        - Implementing a web server
        - Processing request, response, data
    - Storing data in databases
        - Models
        - Object Relational Mapper

        - Interaction with DBs
    - API versioning and documentation

- GraphQL
    - Schemas and Types
    - Thinking in Graphs
    - Serving over HTTP
    - Implementing the GraphQL API
    - Validation and Execution
- gRPC
    - Service Definition- Protobuf
    - Architecture
    - Channels Streaming and Types

Module 5: Securing Application

- Basic Authentication
- API Authorization
- JSON Web Tokens
- OAuth
- SSL, TLS and HTTPS
- Common Vulnerabilities

Module 6: Understanding Frontend Development

- Designing and Structuring Webpages
- Making pages Interactive with JavaScript
- Using Browser based Web APIs- DOM, Web storage
- Client-side JavaScript Frameworks: Features and Advantage, MEAN, MERN
- Implementing Single Page Applications using JavaScript Tech stack
    - The Node Ecosystem
    - Project Setup
    - Creating and styling components
    - Managing Component hierarchies and lifecycle
    - Managing State
    - Routing
- Building, deploying and Hosting

Module 7: Testing

- API Testing
    - API Testing and Types
    - Unit Testing
    - Contract Testing

- Frontend
    - Unit testing
    - Cross browser testing
    - Acceptance Testing

Module 8: Accessibility and Performance

- Accessibility
    - Inclusive Design
    - Assistive Technologies
    - Web content accessibility Guidelines
    - Optimizing Websites for Accessibility
    - Testing Accessibility
- Performance
    - Tools and Metrics for Measuring Performance
    - Options for Optimization
        - Caching- Client side, Server side
        - Minifying Code, Compressing files
        - Lazy Loading

Module 9: Latest Advancements

- Progressive Web Apps
- Web Assembly
- Microfronends

Learning Outcomes:

No	Learning Outcomes
LO1	Understand the underlying architecture used for Web applications and identify the various components of the Web Application
LO2	Demonstrate the creation of API to accomplish various backend functionalities of an application like database interaction, handling user requests
LO3	Design and develop user-friendly and interactive web frontends.
LO4	Implement a functional end-to-end web application using client-side and server-side web technologies.

Part B: Contact Session Plan

Academic Term	
Course Title	**Full Stack Application Development**
Course No	SE* ZG503
Lead Instructor	

Glossary of Terms

1. Contact Hour (CH) is an hour-long live session with students conducted either in a physical classroom or enabled through technology. In this model of instruction, instructor-led sessions will be for 22 CH.
    a. Pre CH = Self-Learning done prior to a given contact hour
    b. During CH = Content to be discussed during the contact hour by the course instructor
    c. Post CH = Self Learning done post the contact hour
2. Contact Session(CS) is a two-hour live session with students conducted either in a physical classroom or enabled through technology. In this model of instruction, instructor led sessions will be for 11 CS.
    a. Pre CS = Self-Learning done before a given contact session
    b. During CS = Content to be discussed during the contact session by the course instructor
    c. Post CS = Self Learning done post the contact session
3. RL stands for Recorded Lecture or Recorded Lesson. It is presented to the student through an online portal. A given RL unfolds as a sequences of video segments interleaved with exercises
4. SS stands for Self-Study, which is done as a study of relevant sections from textbooks and reference books. It could also include study of external resources.
5. LE stands for Lab Exercises
6. HW stands for Home Work.
7. M stands for module. Module is a standalone quantum of designed content. A typical course is delivered using a string of modules. M2 means module 2.

Teaching Methodology (Flipped Learning Model)

The pedagogy for this course is centered around flipped learning model in which the traditional class-room instruction is replaced with recorded lectures to be watched at home as per the student's convenience and the erstwhile home-working or tutorials become the focus of classroom contact sessions. Students are expected to finish the home works on time.

Contact Session Plan

- Each Module (M#) covers an independent topic and module may encompass more than one Recorded Lecture (RL).
- Contact Sessions **(2hrs each week)** are scheduled alternate weeks after the student watches all Recorded Lectures (RLs) of the specified Modules (listed below) during the previous week
- In the flipped learning model, Contact Sessions are meant for in-classroom discussions on cases, tutorials/exercises or responding to student's questions/clarification--- may encompass more than one Module/RLs/CS topic.

- Contact Session topics listed in course structure (numbered CSx.y) may cover several RLs; and as per the pace of instructor/students' learning, the instructor may take up more than one CS topic during each of the below sessions.

Course Contents

Contact Session 1

M1: Application Development- Introduction

Time	Type	Description	Content Reference
Pre-CS	RL	Web Application Cloud Native application Serverless applications Mobile applications Cross Platform Applications Evolution of Web application architectures Monolitic and Microservices application Application Programming Interfaces MVC Pattern	R3- Chapter 12, R1 Web references
During CS	CS1.1	Modern Application Landscape - Comparison	
	CS1.2	Challenges and Practices	
Post-CS	LE1.1		
	SS1.1	Explore which of the Architectural Principles are applied in a given software system	
	HW1.1		

Contact Session 2

M2: Understanding the Basic Web application

Time	Type	Description	Content Reference
Pre-CS	RL	Client-server Architecture Frontend and Backend components Interaction between the components Web servers Domain Names Role of Content Delivery Networks Application Development Tech Stacks	R3- Chapter 1, https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/How_the_Web_works
During CS	CS2.1	Relationship between URLs, Domains, and IP Addresses Technologies and Tools for Full stack development	
	CS2.2	Using Browser's Developer Tools to understand flow of communication	
Post-CS	LE2.1		
	SS2.1		
	HW2.1	Explore the Use of CDN in Caching and Video Streaming	

Contact Session 3

M3: Web Protocols

Time	Type	Description	Content Reference
Pre-CS	RL	HTTP Request- Response and its structure HTTP Methods HTTP Headers Connection management - HTTP/1.1 and HTTP/2 AJAX, Fetch API HTTP Polling Webhooks Server Sent Events Web sockets Message Queues	T1- Chapter 6,8 R3- Chapter 5 https://developer.mozilla.org /en-S/docs/Web/HTTP / Overview
During CS	CS3.1	Demonstration of HTTP request response Demonstration of API calls from frontend Compare HTTP/1 and HTTP /2	
	CS3.2	Demonstration of Webhooks Demonstrate the concepts of Websockets, different types of Message queues	
Post-CS	LE3.1		
	SS3.1		
	HW3.1	Try opening any website and observe the request response through the developer tools	

Contact Session 4

M4: Server Side: Implementing Web Services

Time	Type	Description	Content Reference
Pre-CS	RL	Principles of REST REST constraints REST Maturity Model Interaction Design with HTTP Interface Design (URI) Representation and Metadata design Frameworks- Features and Advantages URL Mapping and Routing Processing Requests and Response Interaction with DBs API Documentation API Versioning API Design Practices	R4- Chapter 4,6 T1- Chapter 13,14
During CS	CS4.1	REST Design for a Choosen Problem Statement Create REST service to perform CRUD operations with the database, using JSON data format.	
	CS4.2	API Documentation- Create Documentation for the REST APIs using Swagger. OpenAPI	
Post-CS	LE4.1		
	SS4.1		
	HW4.1	For a given application, identify the services and draft a REST service Design	

Contact Session 5

M4: Server Side: Implementing Web Services

Time	Type	Description	Content Reference
Pre-CS	RL5	Problems with REST Schemas and Types Thinking in Graphs Serving over HTTP Implementing GraphQL API gRPC-Architecture Service Definition- Protobuf Streams, Messages and Frames Communication Patterns	R4- Chapter 5, 8 R5-Chapter 1, 2,3
During CS	CS5.1	GraphQL- Implementaion Querying gRPC- Demo of gRPC client and server with Protobuf	
	CS5.2	Comparison - Advantage and Disadvantages of API architecture Styles REST vs GRPC vs GraphQL	
Post-CS	LE5.1	Create a REST service, Invoke a service from a web client.	
	SS5.1		
	HW5.1	Compare the advantages of Asynchronous communication.	

Contact Session 6

Module 5: Securing Application

Time	Type	Description	Content Reference
Pre-CS	RL	Basic Authentication API Authorization JSON Web Tokens Oauth OpenID HTTPS Common Vulnerabilites Cross Origin Resource Sharing	T1- Chapter 18 R4- Chapter 11
During CS	CS6.1	Discuss the Various Authentication methods. Implementation scenarios for Oauth	
	CS6.2	Demo to Add Authentication with JWT	
Post-CS	LE6.1		
	SS6.1	Alternative ways for security	
	HW6.1		

Contact Session 7

Module 6: Understanding Frontend Development

Time	Type	Description	Content Reference
Pre-CS	RL	Designing and Structuring Webpages Making pages Interactive with JavaScript Asynchronous Programming Client Side Web APIs Need and Advantages of Frontend Frameworks Comparative study of frontend frameworks The Node Ecosystem Project Setup Building and Deployment	https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Client-side_JavaScript_frameworks https://developer.mozilla.org/en-US/docs/Learn/Tools_and_testing/Client-side_JavaScript_frameworks /React_getting_started
During CS	CS7.1	Setup for Frontend development	
	CS7.2	Event Handing, Repaint and Reflow in Javascript	
Post-CS	LE7.1		
	SS7.1	Simple exercises in Javascript	
	HW7.1		

Contact Session 8

Module 6: Understanding Frontend Development

Time	Type	Description	Content Reference
Pre-CS	RL	React -Introduction React Components JSX Props and State Conditional Rendering React Hooks React Routing Redux	https://react.dev/learn
During CS	CS8.1	Create React components Fetch data from Backend Build React app and Deploy	
	CS8.2	Demonstration of end to end application	
Post-CS	LE8.1		
	SS8.1		
	HW8.1		

Contact Session 9

Module 7: Testing

Time	Type	Description	Content Reference
Pre-CS	RL	API Testing and Types Unit Testing Contract Testing Unit Testing Cross Browser Testing Automated Testing	R4- Chapter 12
During CS	CS9.1	API testing and its types	
	CS9.2	Unit testing demonstration- Jest, Chai API Testing with Postman	
Post-CS	LE9.1		
	SS9.1	Automated Testing and its benefits	
	HW9.1		

Contact Session 10

Module 8: Accessibility and Performance

Time	Type	Description	Content Reference
Pre-CS	RL	Inclusive Design Web content accessibility Guidelines Optimizing Websites for Accessibility Testing Accessibility Tools and Metrics for Measuring Performance Options for Optimization	R3-Chapter 8, 21
During CS	CS10.1	Caching- Client side, Server side Minifying Code, Compressing filesLazy Loading	
	CS10.2	Exercises to improve performance, test accessibility	
Post-CS	LE10.1		
	SS10.1	Accessibility guidelines	
	HW10.1		
	QZ10.1		

Contact Session 11: Latest Advancements and Review Session

Labs:

- Create a REST service to perform CRUD operations with the database, using JSON data format.
- Set up an API gateway which serves as entry point for all services, and to apply security.
- Building a Web-Service Client: Create a service that monitors stock data real-time and consume that API from the frontend application and display the stock price on the page.
- Create a frontend application using React for the given userstories.

Evaluation Scheme

Evaluation	Name	Type	Weight	Duration	Day, Date, Session,

Component	(Quiz, Lab, Project, Midterm exam, End semester exam, etc)	(Open book, Closed book, Online, etc.)			**Time**
EC – 1	Quiz 1		5%		To be announced
	Quiz 2		5%		
	Assignment 1		10%		
	Assignment 2		10%		
EC – 2	Mid-term Exam	Closed book	30%	2 hours	To be announced
EC – 3	End Semester Exam	Open book	40%	3 hours	To be announced

Note *- Evaluation components can be tailored depending on the proposed model.*

Important Information

Syllabus for Mid-Semester Test (Closed Book): CS 1 to CS5
Syllabus for Comprehensive Exam (Open Book): All topics given in plan of study

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Announcements regarding the same will be made in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
3. For Open Book exams: Use of prescribed and reference text books, in original (not photocopies) is permitted. Class notes/slides as reference material in filed or bound form is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam. The genuineness of the reason for absence in the Regular Exam shall be assessed prior to giving permission to appear for the Make-up Exam. Make-Up Test/Exam will be conducted only at selected exam centers on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the handout.